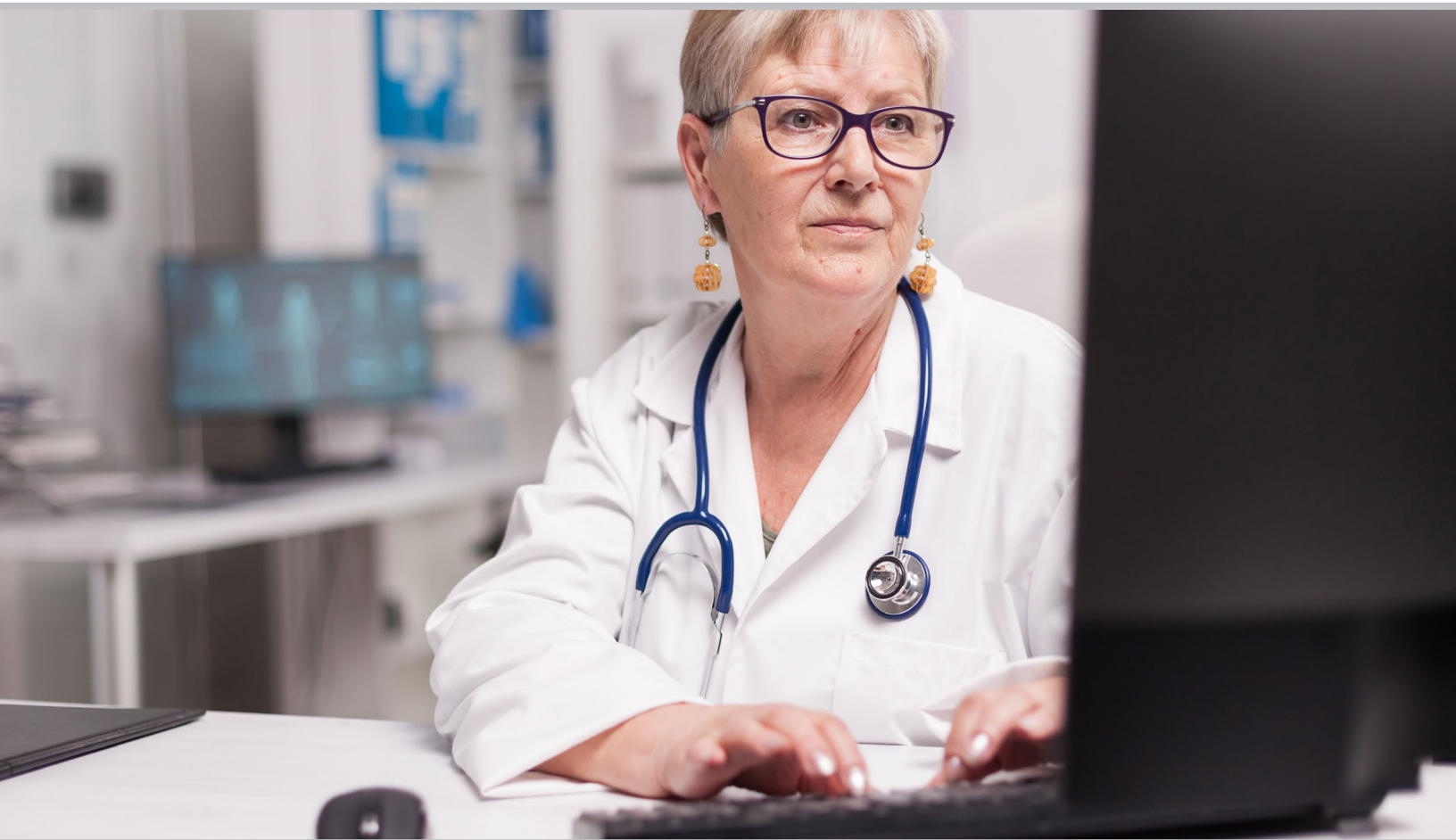




Cisco Catalyst SD-WAN Adoption and Onboarding Guide

A field-ready guide for network engineers deploying, managing, and scaling secure software-defined WAN (SD-WAN) connectivity across the enterprise.

This guide covers:

- SD-WAN architecture fundamentals
- Core components and capabilities
- Deployment models and environments
- WAN Edge onboarding and connectivity
- Template-based device management
- Security, policy, and automation workflows
- Best practices for scaling and maintaining Cisco SD-WAN environments

Simplify operations, strengthen security, and deliver reliable connectivity with Cisco Catalyst SD-WAN.

The Evolution of Modern Networks

Today's enterprise networks are under more pressure than ever—expected to be faster, smarter, and more secure. Before we dive into Cisco Catalyst Software-Defined Wide Area Network (SD-WAN) let's take a moment to understand why legacy WANs no longer meet the demands of a global business.

Watch the video below to explore what's driving the shift toward modern, software-defined networks.



Exploring the Shift to Cisco Catalyst SD-WAN

Legacy WAN architectures were designed for a time when most traffic flowed between branch offices and centralized data centers. However, today's traffic patterns are far more dynamic, often heading toward cloud applications such as Microsoft 365, Webex, Salesforce, and custom logistics platforms. The rigid structure of traditional WANs can't adapt easily to this model, introducing bottlenecks, higher costs, and reduced performance.

Cisco SD-WAN offers a modern, software-defined solution that replaces these hardware-bound limitations with a secure and scalable virtual overlay.

This enables consistent, policy-based connectivity across all network edges, regardless of the transport method.

Core Architecture Elements

Cisco Catalyst SD-WAN architecture is built on a set of key components. These components work together to establish a secure separation between the control and data planes, centralize management, and support automation through open APIs.

Component	Capability
WAN Edge	Connects branch locations to the SD-WAN fabric, forwards data traffic, and enforces local policies.
vManage	Provides centralized configuration, orchestration, monitoring, and analytics for the entire SD-WAN network.
vSmart	Distributes routing and policy information and controls the SD-WAN control plane for consistent policy enforcement.
vBond	Authenticates devices and orchestrates their initial secure connections within the SD-WAN overlay.
vAnalytics	Gathers telemetry and analyzes performance data to deliver operational insights and support capacity planning.

Key Benefits

Cisco Catalyst SD-WAN is purpose-built to support a future defined by cloud-first infrastructure, mobility, and operational agility.

Centralized Orchestration	Enables simplified policy management, provisioning, and security enforcement.
Application-Aware Routing	Centralized enforcement of security and access rules (role, device type, location).
Enhanced Security	Protects data across the network with integrated security features.
Operational Efficiency	Reduces manual effort and provides real-time analytics.
Scalability	Supports rapid growth—essential for a logistics-driven enterprise.

Choosing the Deployment Model

Once the Cisco Catalyst SD-WAN architecture and its core components are understood, the next step is to determine how and where to deploy those components to support your organization's needs best. Cisco Catalyst SD-WAN is highly flexible, offering multiple deployment models that can align with varying operational requirements, scalability goals, and security policies.

Deployment Environments

While SD-WAN controllers can be deployed in a variety of environments, FedEx currently uses an on-premises deployment model for its SD-WAN infrastructure. Outlined below are the available options for context, but day-to-day work will primarily involve the on-prem setup.



On-Premises Data Centers

Controllers can be hosted in private data centers. This option provides direct control over infrastructure, security, and compliance.



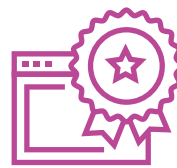
Public Cloud

Controllers can be deployed in AWS, Microsoft Azure, and other cloud environments. This option provides scalability and flexibility.



MSP Deployments

A Managed Service Provider (MSP) can manage deployment, monitoring, and ongoing operations, which allows organizations to offload day-to-day administration.



Enterprise CA Integration

Integration with enterprise certificate authorities (CAs) supports device authentication. It also simplifies certificate management.

Deploying Cisco WAN Edge Devices

At the heart of every branch and remote location in a Cisco Catalyst SD-WAN network is the WAN Edge device. These devices are responsible for securely routing traffic, enforcing policy, and maintaining communication between the site and the SD-WAN fabric. WAN Edge devices form the critical bridge between local operations and its global digital infrastructure.

The Edge Device Deployment Process



Onboard the Device

Register and onboard the device to the SD-WAN fabric.



Apply Credentials

Install and validate the required certificates and credentials.



Establish Secure Tunnels

Form encrypted tunnels with the SD-WAN control plane.



Deployment Complete

The device is connected securely to the SD-WAN fabric.



Important Consideration

Engineers must also account for Network Address Translation (NAT) environments, which can introduce challenges in reachability and device authentication.

Secure onboarding establishes trust throughout the SD-WAN fabric.

Device Configuration at Scale

Template-Based Management

In small networks, manually configuring each device might be manageable. But in a large enterprise, manual configuration becomes inefficient and prone to errors. Cisco Catalyst SD-WAN solves this challenge with a template-based configuration model, designed to streamline provisioning and promote consistency across the network.

Engineers work within a centrally maintained template structure, supported by in-house automation. This ensures consistent configurations across environments and minimizes the need for manual adjustments. While most team members don't build templates themselves, understanding the underlying components can provide valuable insight into how SD-WAN environments are designed and managed.



Configuration Groups

Organize devices by role or location.



Feature Profiles

Define settings for specific functions like VPN, QoS, or routing.



Device & Feature Templates

Combine profiles and apply them to devices to enforce consistent behavior.

Benefits of Templates



Consistent configurations across sites



Reduced manual configuration errors



Faster provisioning of new devices



Simplified operational management



**Cisco Catalyst SD-WAN is
the foundation for secure,
reliable connectivity.**



Bringing It All Together

Cisco Catalyst SD-WAN enables secure, reliable connectivity across distributed healthcare environments by combining centralized management, intelligent traffic routing, and policy-based network control. These capabilities help ensure that clinicians, staff, applications, and critical services remain connected while maintaining the performance, visibility, and security required to support patient care.

Understanding the SD-WAN architecture, deployment options, WAN Edge devices, and configuration management provides the foundation for successful implementation and long-term operations. Together, these components create a network that is scalable, resilient, and prepared to support evolving business and technology needs.

Additional Resources

Use these resources to deepen your understanding of Cisco Catalyst SD-WAN and support ongoing operations:

- [Cisco Catalyst SD-WAN](#): Product information, architecture overviews, feature details, and deployment guidance.
- [Cisco Catalyst SD-WAN Solution Overview](#): Business and technical insights into how Cisco Catalyst SD-WAN delivers secure, scalable connectivity.
- [Cisco Catalyst SD-WAN and Zero Trust](#): Learn how Cisco Catalyst SD-WAN supports Zero Trust principles through secure access, segmentation, and policy enforcement
- [Cisco Learning Network](#): Community discussions, technical articles, and peer-to-peer knowledge sharing.

